

ELT-ESE-3 DSDL

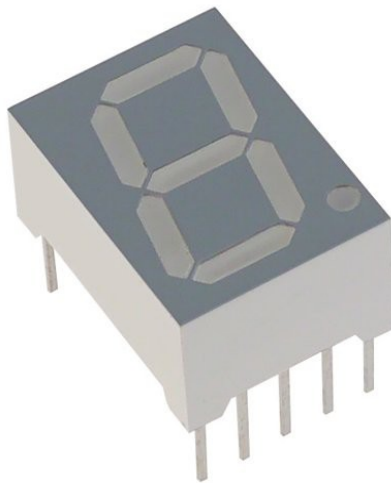


Digital System Design
practical work

HAN Embedded Systems Engineering

E.J Boks MSEE MA

Assignment 1: Design a seven segment driver



Goal:

Design and implement a seven segment driver

Time:

1 week.

Requirements:

- Altera/Intel Quartus.
- TerAsic DE10-Lite board

Description:

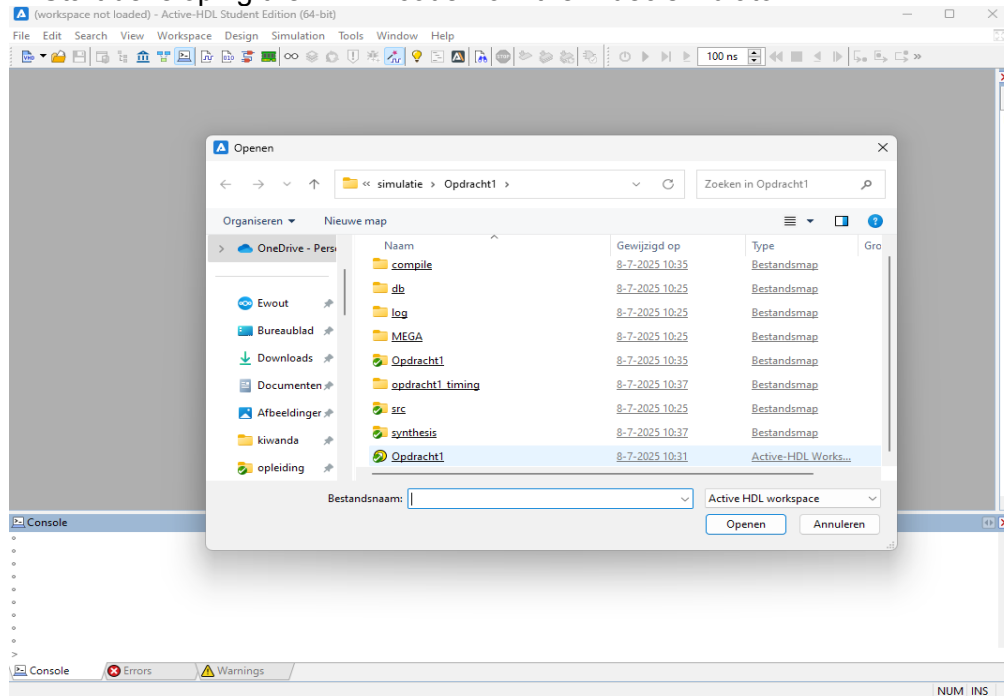
The seven segment displays on the development board can be controlled with a uniform driver. In this assignment you will make, test and demonstrate this driver.

The assignment

Find out how you can control the 7-segment displays (see Chapter 12 of Pedroni 2nd edition / Appendix E 3rd edition).

Study the DE10-Lite manual (paragraph 3.4 DE10-Lite User Manual), which indicates exactly where the seven segment displays are located as seen from the FPGA.

1. Start developing the VHDL code from the Aldec simulator:

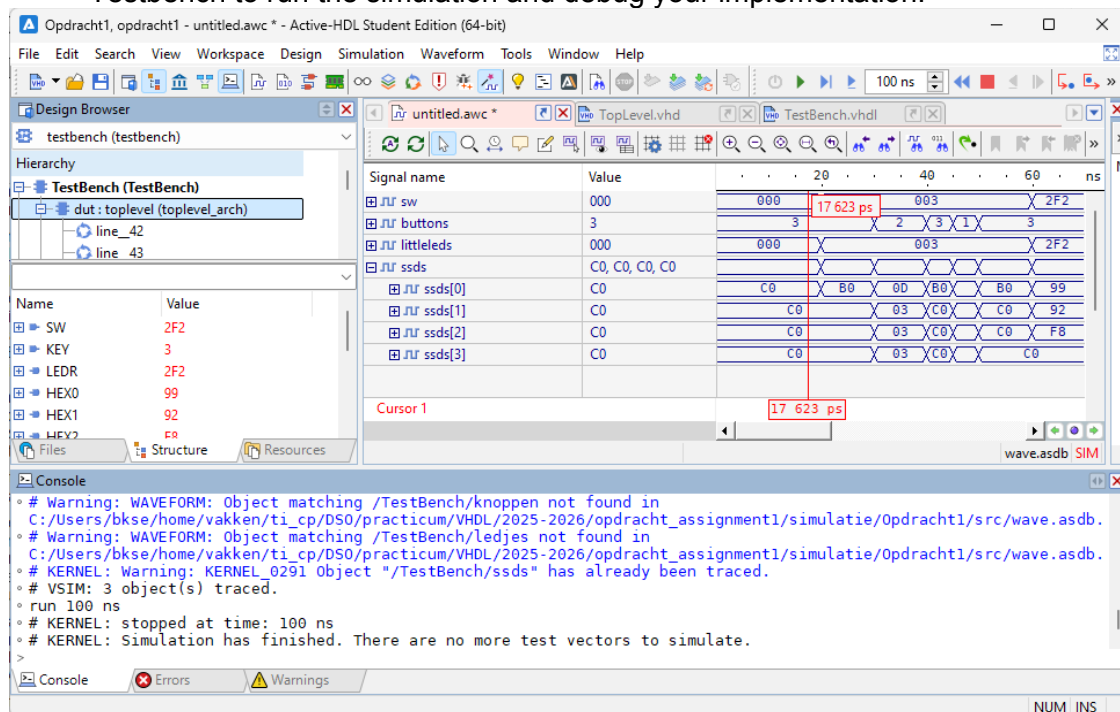


Open the Aldec Workspace "Opdracht 1" in the simulation folder.

The simulator opens, and the file "DE10_Segmentdriver.vhdl" which must be edited is in the project folder. A warning/error will appear where you need to add or modify VHDL code.

2. To get started with Aldec Active HDL, it is useful to first watch [the introductory video](#).

3. Consult paragraph 12.2/Appendix E from Pedroni's book to learn how to implement a *function* within a package that can control a seven-segment display. Please note that the values in the book do not work for the board used by us (DE10-Lite). Please consult the TerAsic board manual for the pin assignments for your particular board.
4. Create in "DE10_7SegmentDriver.vhdl" a driver that can display the numbers 0 .. 9 on a seven segment display, using the VHDL case statement (paragraph 6.7/12.5) Please note that not all characters are easily displayed - use your creativity to come up with a solution!
5. The top level file contains test code and is already completed, nothing needs to be done about there.
6. Train yourself to perform all development work in Aldec. Aldec is very convenient and fast for VHDL development, and the internal simulator is indispensable for testing and debugging an implementation. Use the pre-installed and pre-written Testbench to run the simulation and debug your implementation:



7. If the driver works correctly in the simulation, it's time to switch to Quartus. Development in Quartus is slower, but we'll use Quartus to convert the VHDL file into a programming file for the MAX10.
8. The Quartus project (*.qpf) is already available in the assignment_assignment1 directory, containing all the necessary files.
9. Compile and synthesize the design in Quartus. If everything is correct, you don't need to do anything else.
10. Program and test the design with the board. A binary number on the switches should therefore result in a number on the displays.
11. Show a working demo to the teacher.

Delivery

Show a working application to the teacher.